

Last name: _____
First name: _____
Matricola: _____
M.Sc. program: _____

Politecnico di Milano

Masters of Science in Computer Science and Engineering; Geoinformatics Engineering; Music and Acoustic Engineering; and Telecommunication Engineering

FOUNDATIONS OF OPERATIONS RESEARCH

Exam of XXX XX, 2024

Time available: 2 hours and 10 minutes

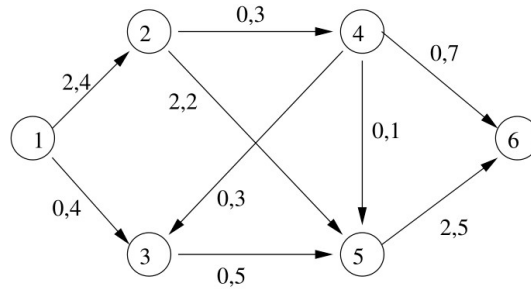
Prob. 1	Prob. 2	Prob. 3	Prob. 4	Prob. 5	Total
7.5	8.5	7.5	3.0	4.5	31

Remarks:

- Write the answers to the various questions in the specified locations **using a blue or black pen**.
- Report and explain all steps on the **back of the previous page**.
- The answers should be as clear and precise as possible.
- No course material or summary is allowed, except for Problem 5 which is open notes/book.

Problem 1

Consider the following network where the first number associated to each arc (link) corresponds to the initial amount of flow (in Gbit/s) crossing it and the second number to the arc capacity (in Gbit/s).



Consider the determination of a feasible flow of maximum value from node 1 to node 6, starting from the given feasible flow of value 2.

a) Which algorithm can be used to solve this problem? Apply it starting from the given feasible flow and report all the intermediate steps.

b) Draw the network and indicate the maximum feasible flow you have found, specifying the amount of flow through each arc and the corresponding total value of the maximum feasible flow.

c) Indicate a cut separating the source (node 1) from the sink (node 6) of minimum capacity on the network of point b).

d) Consider a generic network with n nodes and m arcs. What is the worst-case complexity of the above algorithm when arbitrary augmenting paths are selected at each iteration? Provide a detailed explanation on the back of the **previous page**.

Problem 2

Consider the linear programming problem:

$$\begin{array}{llllll} \max & -6x_1 & -8x_2 & +6x_3 & & \\ \text{s. t.} & -x_1 & -2x_2 & -x_3 & \leq & 1 \\ & 3x_1 & -4x_2 & +3x_3 & \leq & 1 \\ & x_1, & x_2, & x_3 & \geq & 0 \end{array}$$

a) Write the problem in the standard form and indicate a basic solution associated with it. Comment on the feasibility of this solution.

b) Apply one iteration of the simplex method with Bland's rule to this linear program, starting from the solution $\underline{x} = (0, 0, 0)^T$. Report all the steps, including the initial tableau and the resulting tableau. Check whether the obtained solution is optimal and motivate your answer.

c) Write the dual of the original problem.

Note: Part d) below (marked with **Quiz substitute**) replaces the score obtained in the quizzes. Answering it implies neglecting that score.

d) **Quiz substitute** Derive the optimal solution of the dual problem from the optimal solution $\underline{x}^* = (0, 0, 1/3)^T$ of the given linear program. Clearly state the result(s) that justifies(y) your answer.

Problem 3

Consider the following integer linear programming (ILP) problem:

$$\begin{array}{ll} \min & x_1 - 3x_2 \\ \text{s. t.} & -2x_1 + 5x_2 \leq 10 \\ & 6x_1 + 5x_2 \leq 30 \\ & x_1, x_2 \geq 0 \text{ integer} \end{array}$$

a) Describe on the back of the **previous page** how the linear programming relaxation is used in the Branch-and-Bound method for solving ILP problems.

b) How many fractional Gomory cuts can be derived from the optimal solution of the linear programming relaxation of the given ILP problem? Select the most suitable answer.

- i. 1 ii. 2 iii. 4 iv. None of the remaining

c) Find the optimal solution of the above ILP problem with the Branch-and-Bound method by solving graphically the linear programming relaxation of the subproblems. When it applies consider first the subproblem with the best bound.

Report all steps (the branching tree, the graphical solutions of all subproblems with the corresponding solutions and bounds) on the back of the **previous page**.

d) Indicate the optimal solution found and its objective function value.

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Problem 4

Complete adequately the Python statements below to implement the following modeling components:

- parameter n ,
- sets $I = \{1, \dots, n\}$ and $J \subseteq I$,
- parameter d_{ij} for all $i \in I, j \in J$,

```
import mip
from mip import binary

n = 3
m = 3
I = range(1, n+1)
J = range(1, m+1)
d = [[1, 2, 3, 4], [2, 3, 4, 1], [3, 4, 1, 2], [4, 1, 2, 3]]
```

- declare the model

- the variables $x_i \in \{0, 1\}$ for all $i \in I$, and $y_{ij} \geq 0$ for all $i \in I, j \in J$,

- the following objective function to be minimized:

$$\min \sum_{i \in I, j \in J} d_{ij} y_{ij} + \sum_{i \in I} 100 x_i$$

- the following constraints:

$$\sum_{j \in J: j \neq 1 \wedge j \neq n} y_{ij} \leq x_i \quad \forall i \in I$$

- optimize the model.

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Problem 5

A company has to supply a new type of product to n stores across a given area, and it may use m warehouses to do so.

Let

- p_i denote the maximum capacity of the i -th warehouse, for $1 \leq i \leq m$,
- r_j denote the demand of the j -th store, for $1 \leq j \leq n$,
- f_i denote the fixed cost for using warehouse i , for $1 \leq i \leq m$,
- c_{ij} denote the unit transportation cost from warehouse i to store j , for $1 \leq i \leq m$ and $1 \leq j \leq n$.

Due to truck capacity at most k_{ij} units of product can be transported from warehouse i to store j , for $1 \leq i \leq m$ and $1 \leq j \leq n$.

To avoid excessive fragmentation of the delivery, each store must receive the requested product units from at most 3 warehouses. Moreover, at most 5 warehouses can be used.

The problem is to determine a supply plan which minimizes the total (fixed and transportation) costs, while satisfying all demands and other constraints.

Give an integer linear programming formulation for the problem.

Decision variables:

Objective function:

Constraints: